BOVNAR · V1.1

Bovnar (BVNR) — Units & Currencies Reference

Bovnar (BVNR) v1.1 documentation · Units & Currencies Cheat Sheet · 2026-06-16

> Spec version 1.1 (draft) · 163 physical units · 164 fiat currencies · 50 cryptocurrencies

Contents

1. SI Prefixes

Written as `prefix~base` (mandatory `~` separator). Example: `k~m` = kilometre.

Name	Symbol	Factor	Enum (`si_prefix_id_t`)
quetta	`Q`	10^{30}	`si_quetta`
ronna	`R`	10^{27}	`si_ronna`
yotta	`Y`	10^{24}	`si_yotta`
zetta	`Z`	10^{21}	`si_zetta`
exa	`E`	10^{18}	`si_exa`

Name	Symbol	Factor	Enum (`si_prefix_id_t`)
peta	`P`	10^{15}	`si_peta`
tera	`T`	10^{12}	`si_tera`
giga	`G`	10^{9}	`si_giga`
mega	`M`	10^{6}	`si_mega`
kilo	`k`	10^{3}	`si_kilo`
hecto	`h`	10^{2}	`si_hecto`
deca	`da`	10^{1}	`si_deca`
(none)	—	10^{0}	`si_none`
deci	`d`	10^{-1}	`si_deci`
centi	`c`	10^{-2}	`si_centi`
milli	`m`	10^{-3}	`si_milli`
micro	`µ`	10^{-6}	`si_micro`
nano	`n`	10^{-9}	`si_nano`
pico	`p`	10^{-12}	`si_pico`
femto	`f`	10^{-15}	`si_femto`
atto	`a`	10^{-18}	`si_atto`
zepto	`z`	10^{-21}	`si_zepto`
yocto	`y`	10^{-24}	`si_yocto`
ronto	`r`	10^{-27}	`si_ronto`
quecto	`q`	10^{-30}	`si_quecto`

> `µ` is U+00B5 MICRO SIGN (UTF-8 `0xC2 0xB5`). U+03BC (Greek small letter mu) is **not** accepted. `da` is a two-character prefix: `da~m` = decametre.

Prefix-base ambiguities — resolved by the mandatory `~`:

Bare token	Is a base unit	With `~` becomes prefix
`m`	meter (`bu_meter`)	milli
`d`	day (`bu_day`)	deci
`h`	hour (`bu_hour`)	hecto
`T`	tesla (`bu_tesla`)	tera
`G`	gauss (`bu_gauss`)	giga
`P`	poise (`bu_poise`)	peta
`R`	röntgen (`bu_roentgen`)	ronna

Bare token	Is a base unit	With `~` becomes prefix
`f`	farad (`bu_farad`)	femto
`S`	siemens (`bu_siemens`)	(not a prefix — `S` has no prefix role)

Examples: bare `m` = metre; `m~s` = millisecond. Bare `d` = day; `d~s` = decisecond.

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2. IEC Binary Prefixes

Used **only** on `b` (bit) and `B` (byte). Written as `prefix~base`: `Ki~B` = kibibyte.

Name	Symbol	Factor	Enum (`iec_prefix_id_t`)
kibi	`Ki`	2^{10}	`iec_kibi`
mebi	`Mi`	2^{20}	`iec_mebi`
gibi	`Gi`	2^{30}	`iec_gibi`
tebi	`Ti`	2^{40}	`iec_tebi`
pebi	`Pi`	2^{50}	`iec_pebi`
exbi	`Ei`	2^{60}	`iec_exbi`
zebi	`Zi`	2^{70}	`iec_zebi`
yobi	`Yi`	2^{80}	`iec_yobi`
robi	`Ri`	2^{90}	`iec_robi`
quebi	`Qi`	2^{100}	`iec_quebi`

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3. Prefix Validity Rules

Unit category	SI prefixes	IEC prefixes
All physical units (default)	All 24 allowed	Forbidden
`b` (bit) and `B` (byte)	Only ≥ kilo (`k`, `M`, `G`, …, `Q`)	All 10 allowed
`b` and `B` with sub-kilo SI (`d`, `c`, `m`, `µ`, `n`, `p`, `f`, `a`, `z`, `y`, `r`, `q`, `da`, `h`)	**Forbidden**	—
Currency codes	All 24 allowed	**Forbidden**
Old German units (`bu_pfund` … `bu_scheffel`)	**None** (`si_none` only)	**Forbidden**

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4. Physical Units

> **Symbol** = canonical serialized form (produced on output; accepted on input). **Long forms** = accepted on input only; never produced on output. **Factor** = conversion factor to SI base units unless noted.

4.1 SI Base Units

Symbol	Long forms	Name	Enum
`s`	`sec`, `second`, `seconds`	second	`bu_second`
`m`	`meter`, `metre`, `meters`, `metres`	metre	`bu_meter`
`g`	`gram`, `grams`	gram	`bu_gram`
`A`	`amp`, `amps`, `ampere`, `amperes`	ampere	`bu_ampere`
`K`	`kelvin`, `kelvins`	kelvin	`bu_kelvin`
`mol`	`mole`, `moles`	mole	`bu_mol`
`cd`	`candela`, `candelas`	candela	`bu_candela`

> `g` (gram) is the base symbol; `k~g` = kilogram.

4.2 Named SI-Derived Units

Symbol	Long forms	Name	Enum	SI definition
`Hz`	`hertz`	hertz	`bu_hertz`	s^{-1}
`N`	`newton`, `newtons`	newton	`bu_newton`	$kg \cdot m \cdot s^{-2}$
`Pa`	`pascal`, `pascals`	pascal	`bu_pascal`	$kg \cdot m^{-1} \cdot s^{-2}$
`J`	`joule`, `joules`	joule	`bu_joule`	$kg \cdot m^{2} \cdot s^{-2}$
`W`	`watt`, `watts`	watt	`bu_watt`	$kg \cdot m^{2} \cdot s^{-3}$
`V`	`volt`, `volts`	volt	`bu_volt`	$kg \cdot m^{2} \cdot A^{-1} \cdot s^{-3}$
`Ω`	`ohm`, `ohms`, `Ohm`	ohm	`bu_ohm`	$kg \cdot m^{2} \cdot A^{-2} \cdot s^{-3}$ — U+2126 OHM SIGN; U+03A9 (Greek capital omega) **not** accepted
`F`	`farad`, `farads`	farad	`bu_farad`	$kg^{-1} \cdot m^{-2} \cdot A^{2} \cdot s^{4}$
`C`	`coulomb`, `coulombs`	coulomb	`bu_coulomb`	$A \cdot s$
`S`	`siemens`	siemens	`bu_siemens`	$kg^{-1} \cdot m^{-2} \cdot A^{2} \cdot s^{3}$
`Wb`	`weber`, `webers`	weber	`bu_weber`	$kg \cdot m^{2} \cdot A^{-1} \cdot s^{-2}$

Symbol	Long forms	Name	Enum	SI definition
`T`	`tesla`, `teslas`	tesla	`bu_tesla`	$kg \cdot A^{-1} \cdot s^{-2}$
`H`	`henry`, `henrys`, `henries`	henry	`bu_henry`	$kg \cdot m^{2} \cdot A^{-2} \cdot s^{-2}$
`lm`	`lumen`, `lumens`	lumen	`bu_lumen`	cd·sr
`lx`	`lux`	lux	`bu_lux`	$cd \cdot sr \cdot m^{-2}$
`Bq`	`becquerel`, `becquerels`	becquerel	`bu_becquerel`	s^{-1}
`Gy`	`gray`, `grays`	gray	`bu_gray`	$m^{2} \cdot s^{-2}$
`Sv`	`sievert`, `sieverts`	sievert	`bu_sievert`	$m^{2} \cdot s^{-2}$
`kat`	`katal`, `katals`	katal	`bu_katal`	$mol \cdot s^{-1}$
`rad`	`radian`, `radians`	radian	`bu_radian`	dimensionless (m/m)
`sr`	`steradian`, `steradians`	steradian	`bu_steradian`	dimensionless (m^{2}/m^{2})

4.3 Non-SI Units Accepted with SI

Symbol	Long forms	Name	Enum	Factor / notes
`L`, `l`	`liter`, `litre`, `liters`, `litres`	litre	`bu_liter`	10^{-3} m^{3}
`min`	`minute`, `minutes`	minute	`bu_minute`	60 s
`h`	`hour`, `hours`	hour	`bu_hour`	3600 s
`d`	`day`, `days`	day	`bu_day`	86400 s
`wk`	`week`, `weeks`	week	`bu_week`	604 800 s
`mo`	`month`, `months`	month (Julian)	`bu_month`	2 629 800 s (= 365.25 d / 12)
`fn`	`fortnight`, `fortnights`	fortnight	`bu_fortnight`	1 209 600 s (= 14 d)
`yr`	`year`, `years`	year (Julian)	`bu_year`	31 557 600 s
`°`, `deg`	`degr`, `degree`, `degrees`	degree (angle)	`bu_degree`	π/180 rad — U+00B0
`t`	`tonne`	tonne	`bu_tonne`	10^{3} kg
`bar`	—	bar	`bu_bar`	10^{5} Pa
`eV`	`electronvolt`	electronvolt	`bu_electronvolt`	$1.602176634 \times 10^{-19}$ J
`Da`	`dalton`	dalton	`bu_dalton`	$1.66053906660 \times 10^{-27}$ kg

Symbol	Long forms	Name	Enum	Factor / notes
`au`	—	astronomical unit	`bu_astronomical_unit`	$1.495978707 \times 10^{11}$ m
`ha`	`hectare`	hectare	`bu_hectare`	10^{4} m²

4.4 Imperial & US Customary — Length

Symbol	Long forms	Name	Enum	Factor
`in`	`inch`, `inches`	inch	`bu_inch`	0.0254 m (exact)
`ft`	`foot`, `feet`	foot	`bu_foot`	0.3048 m (exact)
`yd`	`yard`, `yards`	yard	`bu_yard`	0.9144 m (exact)
`mi`	`mile`, `miles`	statute mile	`bu_mile`	1609.344 m (exact)
`nmi`	`nautical_mile`, `nautical_miles`	nautical mile	`bu_nautical_mile`	1852 m (exact)
`Å` (U+212B)	`angstrom`, `angstroms`, Å (U+00C5)	ångström	`bu_angstrom`	10^{-10} m
`ly`	`light_year`, `light_years`	light-year	`bu_light_year`	$9.4607304725808 \times 10^{15}$ m
`pc`	`parsec`, `parsecs`	parsec	`bu_parsec`	$3.085677581491367 \times 10^{16}$ m
`fur`	`furlong`, `furlongs`	furlong	`bu_furlong`	201.168 m (exact)
`fath`	`fathom`, `fathoms`	fathom	`bu_fathom`	1.8288 m (exact)
`thou`	`thou`, `mil`, `mils`	thou	`bu_thou`	25.4×10^{-6} m (exact)
`ch`	`chain`, `chains`	chain (Gunter's)	`bu_chain`	20.1168 m (exact)
`rd`	`rod`, `rods`	rod (pole, perch)	`bu_rod`	5.0292 m (exact)

> `thou` and `mil` are synonyms. Canonical output is `thou`. `mil` does **not** mean milliradian; milliradians are written `m~rad`.

4.5 Imperial & US Customary — Mass

Symbol	Long forms	Name	Enum	Factor
`lb`	`lbs`, `pound`, `pounds`	pound (avoirdupois)	`bu_pound`	0.45359237 kg (exact)
`oz`	`ounce`, `ounces`	ounce (avoirdupois)	`bu_ounce`	0.028349523125 kg (exact)

Symbol	Long forms	Name	Enum	Factor
`gr`	`grain`, `grains`	grain	`bu_grain`	6.479891×10^{-5} kg (exact)
`st`	`stone`, `stones`	stone	`bu_stone`	6.35029318 kg (exact)
`tn_sh`	`short_ton`, `short_tons`	short ton (US)	`bu_short_ton`	907.18474 kg (exact)
`tn_l`	`long_ton`, `long_tons`	long ton (UK)	`bu_long_ton`	1016.0469088 kg (exact)
`oz_t`	`troy_ounce`, `troy_ounces`	troy ounce	`bu_troy_ounce`	0.0311034768 kg (exact)
`ct`	`carat`, `carats`	metric carat	`bu_carat`	2×10^{-4} kg (exact)
`slug`	`slugs`	slug	`bu_slug`	14.593902937 kg
`dr`	`dram`, `drams`	dram (avoirdupois)	`bu_dram`	$1.7718451953125 \times 10^{-3}$ kg (exact)
`dwt`	`pennyweight`, `pennyweights`	pennyweight (troy)	`bu_pennyweight`	$1.55517384 \times 10^{-3}$ kg (exact)

4.6 Temperature

Symbol	Long forms	Name	Enum	Conversion
`°C`, `degC`	`degrC`, `degreeC`, `degreesC`, `celsius`	degree Celsius	`bu_celsius`	K = °C + 273.15 **(affine)**
`°F`, `degF`	`degrF`, `degreeF`, `degreesF`, `fahrenheit`	degree Fahrenheit	`bu_fahrenheit`	K = (°F + 459.67) × 5/9 **(affine)**
`°Ra`, `degRa`	`degrRa`, `degreeRa`, `degreesRa`, `rankine`	degree Rankine	`bu_rankine`	K = °Ra × 5/9 (linear)
`°De`, `degDe`	`degrDe`, `degreeDe`, `degreesDe`, `delisle`	degree Delisle	`bu_delisle`	K = 373.15 − °De × 2/3 **(affine)**
`°N`, `degN`	`degrN`, `degreeN`, `degreesN`, `newton_temperature`	degree Newton	`bu_newton_temp`	K = °N × 100/33 + 273.15 **(affine)**
`°Re`, `degRe`	`degrRe`, `degreeRe`, `degreesRe`, `reaumur`	degree Réaumur	`bu_reaumur`	K = °Re × 5/4 + 273.15 **(affine)**
`°Ro`, `degRo`	`degrRo`, `degreeRo`, `degreesRo`, `romer`	degree Rømer	`bu_romer`	K = (°Ro − 7.5) × 40/21 + 273.15 **(affine)**

> Kelvin (`K`) is in §4.1. `R` is reserved for röntgen (§4.20). `N` alone is newton (§4.2); use `°N` or `degN` for Newton temperature.

4.7 Pressure

Symbol	Long forms	Name	Enum	Factor
`atm`	`atmosphere`, `atmospheres`	standard atmosphere	`bu_atmosphere`	101 325 Pa (exact)
`at`	`atmosphere_technical`	atmosphere technical	`bu_atmosphere_technical`	98 066.5 Pa (= 1 kgf/cm^2)
`mmHg`	—	millimetre of mercury	`bu_mmhg`	133.322387415 Pa
`Torr`	`torr`	torr	`bu_torr`	101 325/760 Pa
`psi`	—	pound-force per square inch	`bu_psi`	6894.757293168361 Pa
`inHg`	`inch_hg`, `inch_mercury`	inch of mercury	`bu_inch_hg`	3386.388645 Pa

> `at ≠ atm`: 1 at = 98 066.5 Pa; 1 atm = 101 325 Pa.

4.8 Energy

Symbol	Long forms	Name	Enum	Factor
`cal`	`calorie`, `calories`	thermochemical calorie	`bu_calorie`	4.184 J (exact)
`Btu`	`btu`, `BTU`	International Table BTU	`bu_btu`	1055.05585262 J
`erg`	`ergs`	erg	`bu_erg`	10^{-7} J (exact)
`thm`	`therm`, `therms`	US therm	`bu_therm`	1.05480400×10^{8} J (exact)
`ft_lb`	`foot_pound`, `foot_pounds`	foot-pound	`bu_foot_pound`	1.3558179483 J

> `BTU` (all-caps) is a valid alias: with no `$` sigil the token is a physical-unit lookup (the currency table is only consulted for `$`-prefixed tokens), and it matches `bu_btu`. `Btu` and `btu` are also accepted.

4.9 Power

Symbol	Long forms	Name	Enum	Factor
`hp`	`horsepower`	mechanical horsepower	`bu_horsepower`	745.69987158227 W

Symbol	Long forms	Name	Enum	Factor
`PS`	`CV`, `metric_horsepower`	metric horsepower	`bu_metric_horsepower`	735.49875 W (exact)

4.10 Force

Symbol	Long forms	Name	Enum	Factor
`lbf`	`pound_force`	pound-force	`bu_pound_force`	4.4482216152605 N
`dyn`	`dyne`, `dynes`	dyne	`bu_dyne`	10^{-5} N (exact)
`kip`	`kips`	kip (kilopound-force)	`bu_kip`	4448.2216152605 N
`kgf`	`kilogram_force`	kilogram-force	`bu_kilogram_force`	9.80665 N (exact)

4.11 Speed & Rotation

Symbol	Long forms	Name	Enum	Factor
`kn`	`knot`, `knots`	knot	`bu_knot`	1852/3600 m/s
`rpm`	—	revolutions per minute	`bu_rpm`	1/60 s^{-1}

4.12 Acceleration

Symbol	Long forms	Name	Enum	Factor
`gn`	`standard_gravity`	standard gravity	`bu_standard_gravity`	9.80665 $m \cdot s^{-2}$ (exact)

4.13 Volume — US Liquid

Symbol	Long forms	Name	Enum	Factor
`gal`	`gallon`, `gallons`	US liquid gallon	`bu_gallon`	$3.785411784 \times 10^{-3}$ m^3 (exact)
`qt`	`quart`, `quarts`	US liquid quart	`bu_quart`	$9.46352946 \times 10^{-4}$ m^3
`pt`	`pint`, `pints`	US liquid pint	`bu_pint`	$4.73176473 \times 10^{-4}$ m^3
`cup`	`cups`	US cup	`bu_cup`	$2.365882365 \times 10^{-4}$ m^3
`gi`	`gill`, `gills`	US gill	`bu_gill`	$1.18294118250 \times 10^{-4}$ m^3
`fl_oz`	`fluid_ounce`, `fluid_ounces`	US fluid ounce	`bu_fluid_ounce`	$2.95735295625 \times 10^{-5}$ m^3
`tbsp`	`tablespoon`, `tablespoons`	US tablespoon	`bu_tablespoon`	$1.47867648 \times 10^{-5}$ m^3
`tsp`	`teaspoon`, `teaspoons`	US teaspoon	`bu_teaspoon`	$4.92892159375 \times 10^{-6}$ m^3
`bbl`	`barrel`, `barrels`	petroleum barrel	`bu_barrel`	0.158987294928 m^3

> `cup` (lowercase) = US cup; `CUP` (uppercase) = Cuban Peso. See §6.

4.14 Volume — UK Imperial

Symbol	Long forms	Name	Enum	Factor
`gal_uk`	`gallon_uk`, `gallons_uk`	imperial gallon	`bu_gallon_uk`	4.54609×10^{-3} m³ (exact)
`qt_uk`	`quart_uk`, `quarts_uk`	imperial quart	`bu_quart_uk`	1136.5225×10^{-6} m³
`pt_uk`	`pint_uk`, `pints_uk`	imperial pint	`bu_pint_uk`	568.26125×10^{-6} m³
`gi_uk`	`gill_uk`, `gills_uk`	imperial gill	`bu_gill_uk`	1.420653125×10^{-4} m³ (exact)
`fl_oz_uk`	`fluid_ounce_uk`, `fluid_ounces_uk`	imperial fluid ounce	`bu_fluid_ounce_uk`	28.4130625×10^{-6} m³

4.15 Volume — US Apothecary & Dry

Symbol	Long forms	Name	Enum	Factor
`fl_dr`	`fluid_dram`, `fluid_drams`	US fluid dram	`bu_fluid_dram`	$3.6966911953125\times10^{-6}$ m³
`minim`	`minims`	US minim	`bu_minim`	$6.16115199218750\times10^{-8}$ m³
`pk`	`peck`, `pecks`	US dry peck	`bu_peck`	$8.80976754172\times10^{-3}$ m³
`bsh`	`bushel`, `bushels`	US bushel	`bu_bushel`	$3.523907016688\times10^{-2}$ m³

> `minim` not `min` (which is the minute).

4.16 Area

Symbol	Long forms	Name	Enum	Factor
`ac`	`acre`, `acres`	acre	`bu_acre`	4046.8564224 m² (exact)
`barn`	`barns`	barn	`bu_barn`	10^{-28} m² (exact)

4.17 Angle

Symbol	Long forms	Name	Enum	Factor
`arcmin`	`arcminute`, `arcminutes`	arcminute	`bu_arcminute`	π/10800 rad
`arcsec`	`arcsecond`, `arcseconds`	arcsecond	`bu_arcsecond`	π/648000 rad

Symbol	Long forms	Name	Enum	Factor
`grad`	`gradian`, `gradians`, `gon`	gradian	`bu_grad`	π/200 rad
`rev`	`turn`, `revolution`, `revolutions`, `turns`	revolution	`bu_revolution`	2π rad

4.18 Digital

Symbol	Long forms	Name	Enum
`b`	`bit`, `bits`	bit	`bu_bit`
`B`	`byte`, `bytes`, `Byte`, `Bytes`	byte	`bu_byte`

4.19 CGS Units

Symbol	Long forms	Name	Enum	SI equivalent
`P`	`poise`, `poises`	poise (dynamic viscosity)	`bu_poise`	0.1 Pa·s
`St`	`stokes`, `stoke`	stokes (kinematic viscosity)	`bu_stokes`	10^{-4} $m^2 \cdot s^{-1}$
`G`	`gauss`	gauss (magnetic flux density)	`bu_gauss`	10^{-4} T
`Mx`	`maxwell`, `maxwells`	maxwell (magnetic flux)	`bu_maxwell`	10^{-8} Wb
`Oe`	`oersted`, `oersteds`	oersted (magnetic field strength)	`bu_oersted`	1000/(4π) A/m
`sb`	`stilb`, `stilbs`	stilb (luminance)	`bu_stilb`	10^4 cd/m^2
`ph`	`phot`, `phots`	phot (illuminance)	`bu_phot`	10^4 lx
`Gal`	`galileo`, `galileos`	galileo (acceleration)	`bu_galileo`	10^{-2} m/s^2

4.20 Radiation

Symbol	Long forms	Name	Enum	SI equivalent
`Ci`	`curie`, `curies`	curie (radioactivity)	`bu_curie`	3.7×10^{10} Bq
`R`	`roentgen`, `roentgens`	röntgen (radiation exposure)	`bu_roentgen`	2.58×10^{-4} C/kg
`rem`	`rems`	rem (dose equivalent)	`bu_rem`	10^{-2} Sv

4.21 Logarithmic

Symbol	Long forms	Name	Enum	Notes
`Np`	`neper`, `nepers`	neper	`bu_neper`	dimensionless; 1 Np = 20/ln(10) dB ≈ 8.686 dB
`dB`	`decibel`, `decibels`	decibel	`bu_decibel`	dimensionless

4.22 Electrical Power

Symbol	Long forms	Name	Enum	Notes
`var`	`vars`	var (volt-ampere reactive)	`bu_var`	reactive power; same SI dim as W
`VA`	`volt_ampere`, `volt_amperes`	volt-ampere	`bu_volt_ampere`	apparent power; same SI dim as W

> `W`, `var`, and `VA` all have SI dimension $kg \cdot m^2 \cdot s^{-3}$. `bvn_units_compatible` returns `true` across them; use `.components[0].base` to distinguish.

4.23 Textile Linear Density

Symbol	Long forms	Name	Enum	Factor
`tex`	—	tex	`bu_tex`	1×10^{-6} kg/m (ISO 1144)
`den`	`denier`, `deniers`	denier	`bu_denier`	1/9 000 000 kg/m

> 9 den = 1 tex.

4.24 Old German Units

No Old German unit accepts any SI or IEC prefix (`bvn_prefix_unit_valid` rejects all non-`si_none` prefixes for `bu_pfund` … `bu_scheffel`). Enum values 348–360.

Mass (metric-compatible)

Symbol	Long forms	Name	Enum	Factor
`Pfd`	`pfund`, `pfunds`	Pfund	`bu_pfund`	0.5 kg (exact)
`Ztr`	`zentner`	Zentner	`bu_zentner`	50 kg (exact)
`dz`	`doppelzentner`	Doppelzentner	`bu_doppelzentner`	100 kg (exact)
`lot`	`lots`	Lot	`bu_lot`	15.625×10^{-3} kg (exact)

Length (historical Prussian)

Symbol	Long forms	Name	Enum	Factor
`prln`	`prussian_line`, `linie`	Prussian line	`bu_prussian_line`	2.17953×10^{-3} m
`prz`	`prussian_zoll`, `zoll`	Prussian Zoll	`bu_prussian_zoll`	2.61544×10^{-2} m
`prf`	`prussian_fuss`, `preussischer_fuss`	Prussian Fuß	`bu_prussian_fuss`	3.13853×10^{-1} m

Symbol	Long forms	Name	Enum	Factor
`elle`	`prussian_elle`, `preussische_elle`	Prussian Elle	`bu_prussian_elle`	6.67160×10^{-1} m
`rute`	`prussian_rute`, `preussische_rute`	Prussian Rute	`bu_prussian_rute`	3.76624 m
`klafter`	`prussian_klafter`	Klafter	`bu_klafter`	1.88312 m
`dt_mi`	`deutsche_meile`, `german_mile`	Geographische Meile	`bu_german_mile`	7420.44 m

Area (historical Prussian)

Symbol	Long forms	Name	Enum	Factor
`morgen`	`prussian_morgen`	Morgen (Prussian)	`bu_morgen`	2553.22 m^2

Volume (historical Prussian)

Symbol	Long forms	Name	Enum	Factor
`schffl`	`scheffel`, `prussian_scheffel`	Scheffel (Prussian)	`bu_scheffel`	54.961×10^{-3} m^3

4.25 Additional Physical Units (361–367)

Length

Symbol	Long forms	Name	Enum	Factor
`ftUS`	`survey_foot`	US survey foot	`bu_survey_foot`	1200/3937 m ≈ 0.30480061 m
`lea`	`league`, `leagues`	League (US statute, 3 mi)	`bu_league`	4828.032 m
`cbl`	`cable`, `cables`	Cable (international, ⅒ nmi)	`bu_cable`	185.2 m
`hand`	`hands`	Hand (4 in)	`bu_hand`	0.1016 m

Mass

Symbol	Long forms	Name	Enum	Factor
`qntl`	`quintal`, `quintals`	Metric quintal	`bu_quintal`	100 kg
`sc`	`scruple`, `scruples`	Apothecary scruple (20 gr)	`bu_scruple`	1.2959782×10^{-3} kg

Signal Rate

Symbol	Long forms	Name	Enum	SI dimension
`Bd`	`baud`, `bauds`	Baud (symbol/s)	`bu_baud`	s^{-1}

SI prefixes are accepted on all units in this section. IEC prefixes are rejected for all non-digital units.

4.26 Ratio and Proportion (372–377)

Dimensionless scaling factors: `5 %` ≡ `0.05`, `250 ppm` ≡ `0.00025`. These do **not** accept SI or IEC prefixes.

Symbol	Long forms	Name	Enum	Factor
`%`	`percent`	per cent	`bu_percent`	10^{-2}
`‰`	`per_mille`	per mille	`bu_per_mille`	10^{-3}
`‱`	`per_myriad`	per myriad	`bu_per_myriad`	10^{-4}
`pcm`	`per_cent_mille`	per cent mille	`bu_per_cent_mille`	10^{-5}
`ppm`	—	parts per million	`bu_ppm`	10^{-6}
`ppb`	—	parts per billion	`bu_ppb`	10^{-9}

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5. Currencies

5.1 The Mandatory Currency Sigil

As of spec 1.0 a currency code carries a **mandatory** `$` **sigil**: write `$USD`, `$BTC`, `k~$EUR`, `$USD/oz_t`. The codes listed in the tables below are the bare ISO 4217 / crypto identifiers — prefix each with `$` when you use it in a document. A bare code (no `$`) is **not** a currency; it is matched against the physical-unit table and raises `error_unit_illegal` if it is not a unit. This removes every currency/unit namespace collision (e.g. `$CUP` the Cuban Peso vs `cup` the unit).

5.2 ISO 4217 Fiat Currencies

164 codes occupying `value_base_unit_t` slots **134 … 297** (`BVN_CURRENCY_FIAT_FIRST … BVN_CURRENCY_FIAT_LAST`). Currencies have no named `bu_*` enumerators — they are resolved from the `$`-sigil code by `bvn_parse_currency_str` and carried as the numeric `base` value; query them with `bvn_unit_is_fiat` / `bvn_currency_info`.

> **Min** = minor unit exponent N: 1 major unit = 10^N minor units (e.g. 1 USD = 100 cents, N=2). Minor units are **bold** when they differ from 2. `numeric_code` is the ISO 4217 numeric identifier.

Code	Num	Min	Name
`AED`	784	2	UAE Dirham

Code	Num	Min	Name
AFN	971	2	Afghan Afghani
ALL	8	2	Albanian Lek
AMD	51	2	Armenian Dram
ANG	532	2	Netherlands Antillean Guilder
AOA	973	2	Angolan Kwanza
ARS	32	2	Argentine Peso
AUD	36	2	Australian Dollar
AWG	533	2	Aruban Florin
AZN	944	2	Azerbaijani Manat
BAM	977	2	Bosnia-Herzegovina Convertible Mark
BBD	52	2	Barbados Dollar
BDT	50	2	Bangladeshi Taka
BGN	975	2	Bulgarian Lev
BHD	48	**3**	Bahraini Dinar
BIF	108	**0**	Burundian Franc
BMD	60	2	Bermudian Dollar
BND	96	2	Brunei Dollar
BOB	68	2	Boliviano
BRL	986	2	Brazilian Real
BSD	44	2	Bahamian Dollar
BTN	64	2	Bhutanese Ngultrum
BWP	72	2	Botswana Pula
BYN	933	2	Belarusian Ruble
BZD	84	2	Belize Dollar
CAD	124	2	Canadian Dollar
CDF	976	2	Congolese Franc
CHF	756	2	Swiss Franc
CLF	990	**4**	Unidad de Fomento
CLP	152	**0**	Chilean Peso
CNY	156	2	Chinese Yuan
COP	170	2	Colombian Peso
CRC	188	2	Costa Rican Colon

Code	Num	Min	Name
`CUP`	192	2	Cuban Peso
`CVE`	132	2	Cape Verdean Escudo
`CZK`	203	2	Czech Koruna
`DJF`	262	**0**	Djiboutian Franc
`DKK`	208	2	Danish Krone
`DOP`	214	2	Dominican Peso
`DZD`	12	2	Algerian Dinar
`EGP`	818	2	Egyptian Pound
`ERN`	232	2	Eritrean Nakfa
`ETB`	230	2	Ethiopian Birr
`EUR`	978	2	Euro
`FJD`	242	2	Fijian Dollar
`FKP`	238	2	Falkland Islands Pound
`GBP`	826	2	Pound Sterling
`GEL`	981	2	Georgian Lari
`GHS`	936	2	Ghanaian Cedi
`GIP`	292	2	Gibraltar Pound
`GMD`	270	2	Gambian Dalasi
`GNF`	324	**0**	Guinean Franc
`GTQ`	320	2	Guatemalan Quetzal
`GYD`	328	2	Guyanese Dollar
`HKD`	344	2	Hong Kong Dollar
`HNL`	340	2	Honduran Lempira
`HRK`	191	2	Croatian Kuna
`HTG`	332	2	Haitian Gourde
`HUF`	348	2	Hungarian Forint
`IDR`	360	2	Indonesian Rupiah
`ILS`	376	2	Israeli New Shekel
`INR`	356	2	Indian Rupee
`IQD`	368	**3**	Iraqi Dinar
`IRR`	364	2	Iranian Rial
`ISK`	352	**0**	Icelandic Krona

Code	Num	Min	Name
`JMD`	388	2	Jamaican Dollar
`JOD`	400	**3**	Jordanian Dinar
`JPY`	392	**0**	Japanese Yen
`KES`	404	2	Kenyan Shilling
`KGS`	417	2	Kyrgyzstani Som
`KHR`	116	2	Cambodian Riel
`KMF`	174	**0**	Comorian Franc
`KPW`	408	2	North Korean Won
`KRW`	410	**0**	South Korean Won
`KWD`	414	**3**	Kuwaiti Dinar
`KYD`	136	2	Cayman Islands Dollar
`KZT`	398	2	Kazakhstani Tenge
`LAK`	418	2	Laotian Kip
`LBP`	422	2	Lebanese Pound
`LKR`	144	2	Sri Lankan Rupee
`LRD`	430	2	Liberian Dollar
`LSL`	426	2	Lesotho Loti
`LYD`	434	**3**	Libyan Dinar
`MAD`	504	2	Moroccan Dirham
`MDL`	498	2	Moldovan Leu
`MGA`	969	2	Malagasy Ariary
`MKD`	807	2	Macedonian Denar
`MMK`	104	2	Myanmar Kyat
`MNT`	496	2	Mongolian Togrog
`MOP`	446	2	Macanese Pataca
`MRU`	929	2	Mauritanian Ouguiya
`MUR`	480	2	Mauritian Rupee
`MVR`	462	2	Maldivian Rufiyaa
`MWK`	454	2	Malawian Kwacha
`MXN`	484	2	Mexican Peso
`MYR`	458	2	Malaysian Ringgit
`MZN`	943	2	Mozambican Metical

Code	Num	Min	Name
NAD	516	2	Namibian Dollar
NGN	566	2	Nigerian Naira
NIO	558	2	Nicaraguan Cordoba
NOK	578	2	Norwegian Krone
NPR	524	2	Nepalese Rupee
NZD	554	2	New Zealand Dollar
OMR	512	**3**	Omani Rial
PAB	590	2	Panamanian Balboa
PEN	604	2	Peruvian Sol
PGK	598	2	Papua New Guinean Kina
PHP	608	2	Philippine Peso
PKR	586	2	Pakistani Rupee
PLN	985	2	Polish Zloty
PYG	600	**0**	Paraguayan Guarani
QAR	634	2	Qatari Riyal
RON	946	2	Romanian Leu
RSD	941	2	Serbian Dinar
RUB	643	2	Russian Ruble
RWF	646	**0**	Rwandan Franc
SAR	682	2	Saudi Riyal
SBD	90	2	Solomon Islands Dollar
SCR	690	2	Seychellois Rupee
SDG	938	2	Sudanese Pound
SEK	752	2	Swedish Krona
SGD	702	2	Singapore Dollar
SHP	654	2	Saint Helena Pound
SLE	925	2	Sierra Leonean Leone
SLL	694	2	Sierra Leonean Leone (old)
SOS	706	2	Somali Shilling
SSP	728	2	South Sudanese Pound
SRD	968	2	Surinamese Dollar
STN	930	2	Sao Tome and Principe Dobra

Code	Num	Min	Name
`SVC`	222	2	Salvadoran Colon
`SYP`	760	2	Syrian Pound
`SZL`	748	2	Swazi Lilangeni
`THB`	764	2	Thai Baht
`TJS`	972	2	Tajikistani Somoni
`TMT`	934	2	Turkmenistan Manat
`TND`	788	**3**	Tunisian Dinar
`TOP`	776	2	Tongan Pa'anga
`TRY`	949	2	Turkish Lira
`TTD`	780	2	Trinidad and Tobago Dollar
`TWD`	901	2	New Taiwan Dollar
`TZS`	834	2	Tanzanian Shilling
`UAH`	980	2	Ukrainian Hryvnia
`UGX`	800	**0**	Ugandan Shilling
`USD`	840	2	US Dollar
`UYU`	858	2	Uruguayan Peso
`UZS`	860	2	Uzbekistani Som
`VES`	928	2	Venezuelan Bolivar Soberano
`VND`	704	**0**	Vietnamese Dong
`VUV`	548	**0**	Vanuatu Vatu
`WST`	882	2	Samoan Tala
`XAF`	950	**0**	CFA Franc BEAC
`XAG`	961	**0**	Silver
`XAU`	959	**0**	Gold
`XCD`	951	2	East Caribbean Dollar
`XDR`	960	**0**	Special Drawing Rights
`XOF`	952	**0**	CFA Franc BCEAO
`XPD`	964	**0**	Palladium
`XPF`	953	**0**	CFP Franc
`XPT`	962	**0**	Platinum
`XTS`	963	**0**	Test (ISO 4217 reserved)
`YER`	886	2	Yemeni Rial

Code	Num	Min	Name
`ZAR`	710	2	South African Rand
`ZMW`	967	2	Zambian Kwacha
`ZWL`	932	2	Zimbabwean Dollar

> `CLF` is the only currency with 4 minor units. `XTS` is the ISO 4217 testing code; present in the table but should not appear in production data.

5.3 Cryptocurrencies

50 codes occupying `value_base_unit_t` slots **298 … 347** (`BVN_CURRENCY_CRYPTO_FIRST … BVN_CURRENCY_CRYPTO_LAST`). Like the fiat codes they have no named `bu_*` enumerators — resolved by `bvn_parse_currency_str`, queried with `bvn_unit_is_crypto` / `bvn_currency_info`. `numeric_code = 0` for all.

> **Min** = `minor_unit` = on-chain decimal places. E.g. `<uint:64,$BTC>` stores satoshis; divide by 10^8 to obtain BTC.

Code	Min	Subunit	Name
`BTC`	8	satoshi	Bitcoin
`ETH`	18	wei	Ethereum
`SOL`	9	lamport	Solana
`XRP`	6	drop	XRP
`BNB`	18	—	BNB
`ADA`	6	lovelace	Cardano
`LTC`	8	—	Litecoin
`DOT`	10	planck	Polkadot
`XMR`	12	piconero	Monero
`ETC`	18	—	Ethereum Classic
`BCH`	8	—	Bitcoin Cash
`XLM`	7	stroop	Stellar
`FIL`	18	—	Filecoin
`ICP`	8	—	Internet Computer
`TRX`	6	—	TRON
`EOS`	4	—	EOS
`VET`	18	—	VeChain
`NEO`	8	—	Neo

Code	Min	Subunit	Name
`ZEC`	8	—	Zcash
`UNI`	18	—	Uniswap
`ARB`	18	—	Arbitrum
`SUI`	9	—	Sui
`TON`	9	—	Toncoin
`INJ`	18	—	Injective
`SEI`	6	—	Sei
`APT`	8	—	Aptos
`TAO`	9	—	Bittensor
`WIF`	6	—	dogwifhat
`DOGE`	8	koinu	Dogecoin
`LINK`	18	—	Chainlink
`USDT`	6	—	Tether
`USDC`	6	—	USD Coin
`AVAX`	18	—	Avalanche
`ATOM`	6	—	Cosmos
`POL`	18	—	Polygon
`NEAR`	24	—	NEAR Protocol
`ALGO`	6	—	Algorand
`HBAR`	8	—	Hedera
`AAVE`	18	—	Aave
`MKR`	18	—	Maker
`DAI`	18	—	Dai
`STX`	6	—	Stacks
`GRT`	18	—	The Graph
`LDO`	18	—	Lido DAO
`BONK`	5	—	Bonk
`PEPE`	18	—	Pepe
`SHIB`	18	—	Shiba Inu
`JUP`	6	—	Jupiter
`PYTH`	6	—	Pyth Network
`RUNE`	8	—	THORChain

5.4 Currency Prefix Rules

All 24 SI prefixes are allowed on all currency units. IEC binary prefixes are forbidden on all currencies (`error_unit_illegal`).

Example	Meaning
`k~USD`	thousands of USD (×10^3)
`M~EUR`	millions of EUR (×10^6)
`G~ETH`	giga-ETH = Gwei scale (×10^9)
`m~USD`	milli-USD = one tenth of a cent (×10^{-3})

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6. Symbol Disambiguation

As of spec 1.0 a currency is written **only** with the `$` sigil (§5.1), so the bare form is always a physical-unit lookup and the namespaces never collide:

Token	Bare form (no `$`)	`$`-sigil form
`cup` / `CUP`	`cup` → US cup (`bu_cup`); `CUP` → `error_unit_illegal`	`$CUP` → Cuban Peso (ISO 4217:192)
`BTU`	`BTU` → BTU (`bu_btu`); `Btu`, `btu` also accepted	(not ISO 4217)
`SOL`	`SOL` → `error_unit_illegal`	`$SOL` → Solana (crypto)
`BAR`	`BAR` → `error_unit_illegal`; use lowercase `bar`	(not ISO 4217)
`ERG`	`ERG` → `error_unit_illegal`; use lowercase `erg`	(not ISO 4217)
`CAD`, `XAU`	`error_unit_illegal` (no physical unit)	`$CAD` → Canadian Dollar; `$XAU` → Gold (X-code)

No bare token is simultaneously a valid physical unit and a currency: currencies live entirely under `$`, physical units entirely without it.

Physical unit enum range: 1–133, 348–367, 368–371, and 372–377 (163 total) · Fiat: 134–297 (164) · Crypto: 298–347 (50) *`BVN_VALUE_BASE_UNIT_COUNT`* = 378 (*`bu_ppb + 1`*)